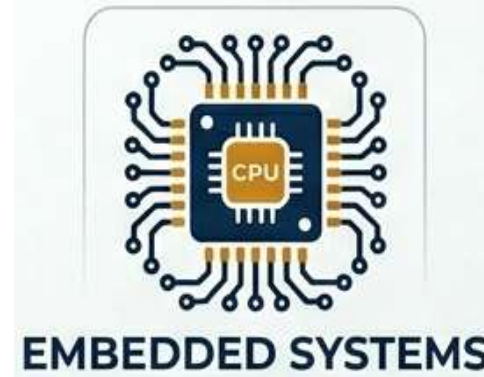




# Automation | Robotics | Customized Solutions | UAV Systems

Light of Innovations & Adoptive Technologies

# Our Presence



# Laser Marking & Cutting Solution



| Parameter                | Description                                          | Remark                                                                                                                                                   |
|--------------------------|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Laser                    | RAYCUS Photonics fiber optic                         |                                                                                                                                                          |
| Laser Power              | 20W                                                  |                                                                                                                                                          |
| Mode of Operation        | Automatic Pulsating                                  |                                                                                                                                                          |
| Max. Output Power        | 900W                                                 |                                                                                                                                                          |
| Source                   | Ytterbium fiber                                      |                                                                                                                                                          |
| Power Tunability         | 10% to 100%                                          |                                                                                                                                                          |
| Pulse Repetition Rate    | 20KHz to 200KHz                                      |                                                                                                                                                          |
| Depth                    | Material dependent / 40um for SS316 in single stroke | To achieve desired depth, required multiple passes. There is no any thumb rule. After 2cd pass, it will start removing 20um as the focal length changed. |
| Operating Temperature    | 0 to 40 Deg. C                                       |                                                                                                                                                          |
| Lens                     | HS Optic                                             |                                                                                                                                                          |
| Wave Length              | 1064                                                 |                                                                                                                                                          |
| Pulse duration @20kHz    | 100ns                                                |                                                                                                                                                          |
| Class (Safety)           | IV                                                   |                                                                                                                                                          |
| Power Stability          | 1 to 3%                                              |                                                                                                                                                          |
| Pulse Energy @20kHz      | 85%/ 80W                                             |                                                                                                                                                          |
| Inbuilt Guide Laser      | WL 450nm, Yes, appro 2W                              |                                                                                                                                                          |
| Beam Quality             | >1.6 m2                                              |                                                                                                                                                          |
| Marking Area             | 100mm*100mm                                          | wave legth & power value exact                                                                                                                           |
| Source Life              | 100000 hrs                                           |                                                                                                                                                          |
| Resolution               | 100um                                                |                                                                                                                                                          |
| Repeatability            | 20um                                                 |                                                                                                                                                          |
| Beam (Ø)                 | 25 to 30 um                                          | cable dia: ID 5mm fiber, 2 mtr cable length                                                                                                              |
| Cooling System           | Active Air cooling                                   |                                                                                                                                                          |
| Power Supply             | 230VAC, 50Hz                                         |                                                                                                                                                          |
| Min Line Width           | 25 to 30um                                           |                                                                                                                                                          |
| Hand Held Unit Weight    | 8 Kg                                                 |                                                                                                                                                          |
| Hand Held Unit Dimension | 100*150*580mm                                        | without focus stand ; Dia 100*160mm focus stand. Column height is 720 mm                                                                                 |



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# AI Machine Vision Bullet Inspection



| Sr.No. | Technical Specification                                                          |
|--------|----------------------------------------------------------------------------------|
| 1      | Conveyor based Inspection system includes bowl feeder, part locator & separator, |
| 2      | Camera for top, bottom & 4 sides of job : 6 cameras                              |
| 3      | 2d laser profiler : 4 nos for measurement                                        |
| 4      | Job feeding mechanism                                                            |
| 5      | lumitrax lighting system:                                                        |
| 6      | Mechanical structure, sensors                                                    |
| 7      | nvidia GPU, screen & peripheral                                                  |
| 8      | Control panel as mentioned in SOR                                                |
| 9      | Designing architecture, Lib                                                      |
| 10     | IP creation for Machine learning database                                        |
| 11     | User interface, PC application & database                                        |
| 12     | ML of 4 components & AI                                                          |
| 13     | Rejection mechanism                                                              |

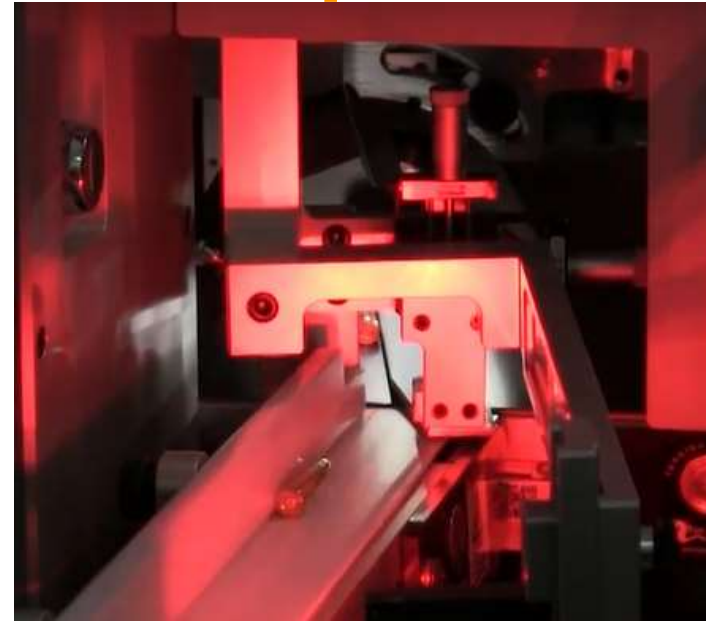
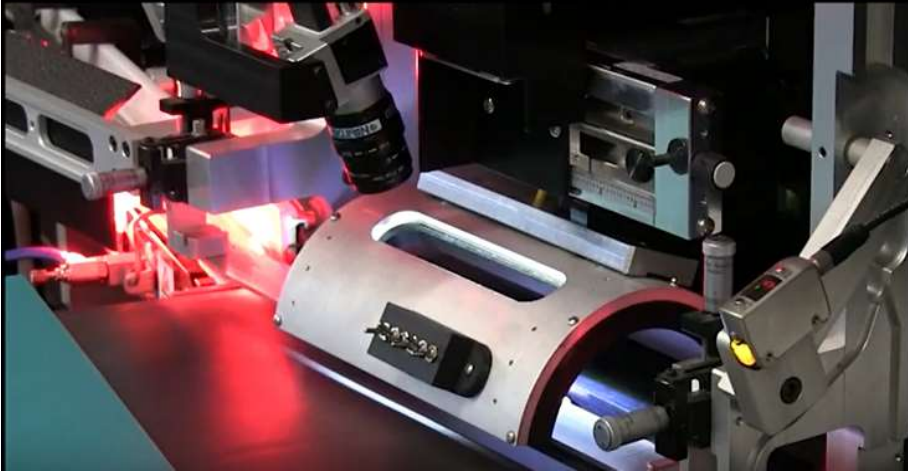


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# AI Machine Vision Bullet Inspection



Surface Camera



360° inspection



Each Laser scans the profile

100,000/sec



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# Automatic Pressure Control System



|    |                               |                  |                |                                                                          |
|----|-------------------------------|------------------|----------------|--------------------------------------------------------------------------|
| 1  | Pressure Range                | 0-1              | psi            | As per the capacity of the pressure sensor                               |
| 2  | Pressure set points           | 0.8              | psi            | As per the capacity of the pressure sensor                               |
|    |                               | 0.6              | psi            | As per the capacity of the pressure sensor                               |
|    |                               | 0.4              | psi            | As per the capacity of the pressure sensor                               |
| 3  | Air Blower motor(Cylindrical) | 1.8KW            | HP             | High Pressure Electric Pump, Model No. HL-8101, Input 220-240VAC, 1800W  |
| 4  | Motor voltage                 | 220-240          | VAC            | As mentioned above                                                       |
| 5  | Phase                         | Single           |                | 210-240VAC                                                               |
| 6  | Load driving                  | Contactor        | 2 O/P REQUIRED | 1. Contactor for motor operation;<br>2. Solenoid valve                   |
| 7  | Input supply available        | 230VSC           |                |                                                                          |
| 8  | Display                       | 2 line/4 line    |                | 3.2" color display                                                       |
| 9  | Resolution                    | 0.01PSI          |                | 0.01 psi                                                                 |
| 10 | Accuracy                      | 1%               |                | ± 1 %                                                                    |
| 11 | Operation                     | Touch            |                | Resistive touch operation                                                |
| 12 | Operating temp.               | 55DEG C          |                | Max 55°C                                                                 |
| 13 | Solenoid Valve                | 0-2 psi          |                | Make: Aira Euro Automation, Model : SS3; Supply: 230VAC; Pressure: 0-2Kg |
| 14 | Enclosure size                | 150X150X100      |                | Controller fitting box                                                   |
| 15 | IP rating if any              | Water/rain proof |                | Controller encloser only                                                 |
| 16 | Additional Accessories        |                  |                | Movable Trolley for controller & blower mounting                         |



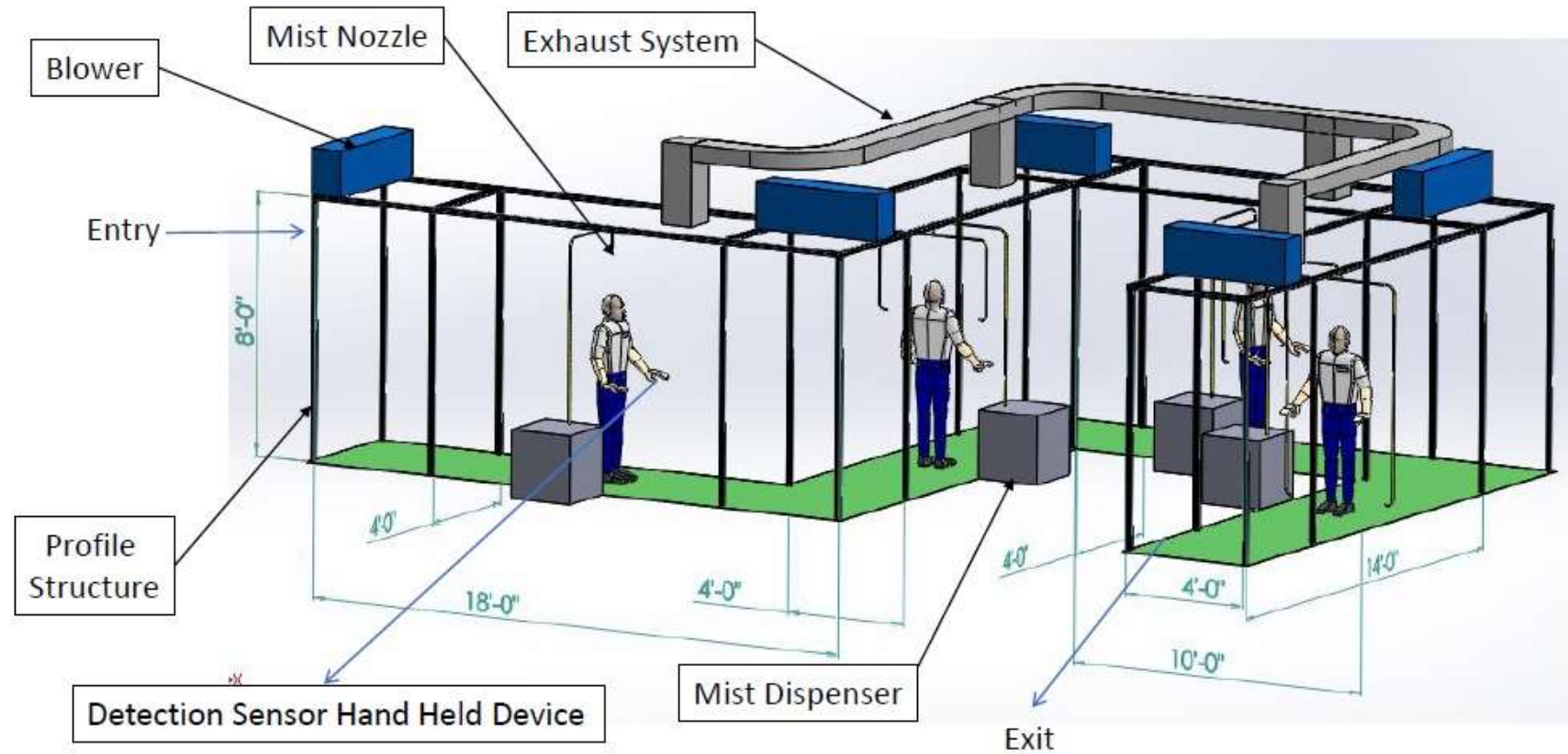
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# CBRN Chemical Test Tunnel



1. Four Chamber Test Tunnel.
2. 10um chemical droplet spraying system.
3. High pressure chemical pumps.
4. Toxic Chemical detection Sensor .
5. Programming of Controller
6. Spraying after Human detection.
7. Programming of HMI.
8. Pre-programmable system for multiple chemical.
9. Suction facility to clean the tunnel after test
10. Power supply 440VAC

# CBRN Chemical Test Tunnel



# CBRN Military Engineering Chemical Test Tunnel



Simulation of Chemicals in Closed Chamber, CME Pune



# DEFENCE DRONES

---

## Core Performance & Capabilities

A high-level overview of the platform's key metrics.

# Key Performance Metrics



## 40-45 Min Endurance

Extended flight time for  
persistent ISR and long-range  
delivery missions.



## 5-35kg Overall Payload

High lift capacity for  
advanced sensors and multi-  
package delivery.



## 5-20 km Range

Long-range LOS operational  
control and reliable, high-  
quality video link.





## Design: Airframe & Propulsion

- > **Airframe:** Compact, foldable design with a customised wheelbase for maximum stability and transportability
- > **Propulsion:** High-thrust-rated motors provide extreme power and operational redundancy.
- > **Power:** Dual 6S 22,000 mAh LiPo pack configuration for reliable, high-discharge power.



# Advanced LiDAR Cam & Datalink

## Real-Time Situational Awareness

The Aetherion H-XX is built for persistent overwatch, providing clear, actionable intelligence.

- \* User-defined live video feed (variable 360p-1080p).
- \* High-bandwidth 4K streaming at ~5km LOS.
- \* Video quality dynamically adapts to link strength.
- \* Multiple tools for Analysis interpretation



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# Other

# Vs

# We



## Feature

## Manual Planning / Basic Apps

## UgCS LiDAR Software

### IMU Calibration

Manual patterns, inconsistent execution

Automated figure-eight and back-and-forth patterns built into route

### Terrain Following

Generic DEM only, fixed altitude risks

Custom DSM import tracks vegetation height, maintains precise AGL

### Mission Planning

Tablet screen, limited visibility on large areas

Desktop interface for 200+ km corridor scans with elevation profile

### Corridor Width Calculation

Manual math for FOV and overlap

Automatic calculation based on sensor specs and altitude

### Offline Operation

Requires internet for maps and elevation data

Map and elevation caching for remote site operations

### Battery Management

Flight stops, manual restart, coverage gaps

Pause/resume from last point after battery swap

### Multi-Sensor Support

Limited to manufacturer payloads

Custom camera profiles for any LiDAR system (Velodyne, RIEGL, etc.)

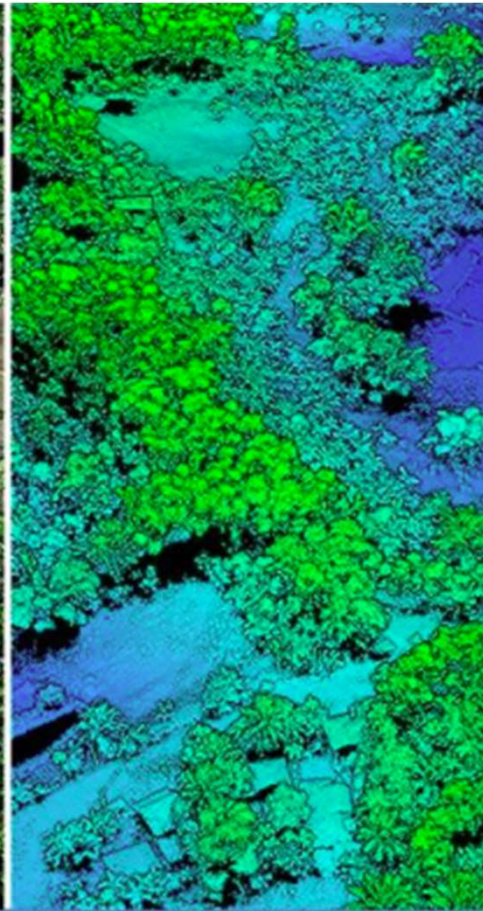
# High Resolution Customised Software



Aerial View



Point Cloud View

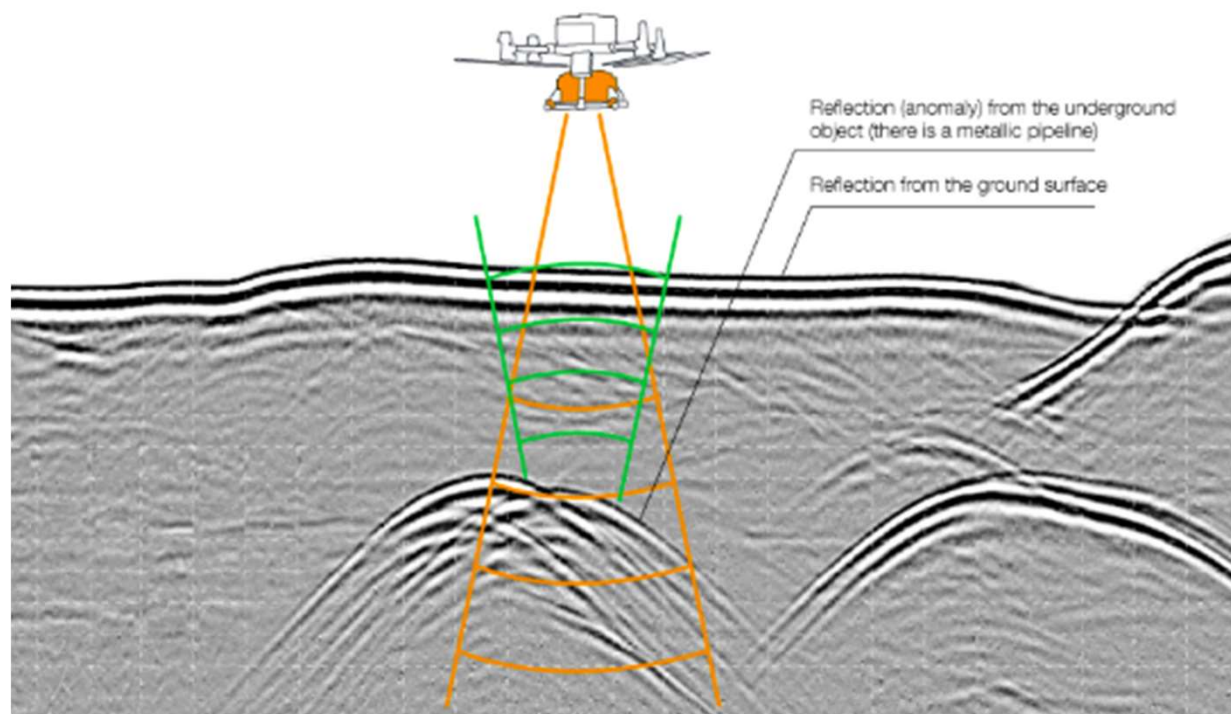


3D DEM

# Ground Penetrating Radar



Ground-penetrating Radar (GPR)  
mounted on a drone



Drone- Mounted GPR Detecting underground  
Anomalies

Please note that penetration and resolution in certain places depend on soil composition, humidity, temperature, etc. In the table below, we used the parameters of a typical "average soil"—some substance with a relative dielectric permittivity of 9, low conductivity, and low water content.

By request, Zond Aero LF GPR systems may come with antennas for custom central frequencies.

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| Central frequency, MHz                                                                                        | 1000                          | 600                                  | 500           | 300          | 150          | 100          |
|---------------------------------------------------------------------------------------------------------------|-------------------------------|--------------------------------------|---------------|--------------|--------------|--------------|
| GPR model                                                                                                     | Zond Aero 1000                | Zond Aero 600 & MALÅ<br>GeoDrone 600 | Zond Aero 500 | Zond Aero LF | Zond Aero LF | Zond Aero LF |
| Penetration from surface, m                                                                                   | 0.5 .. 1                      | 2 .. 4                               | 2 .. 4        | 4 .. 8       | 8 .. 15      | 15 .. 20     |
| Penetration from the drone, m                                                                                 | 0.3 .. 0.5                    | 1 .. 2                               | 1 .. 2        | 2 .. 4       | 4 .. 8       | 7 .. 10      |
| Penetration from the drone in freshwater, m (water conductivity <200 µS/cm)                                   | -                             | 0.25                                 | 0.5           | 2            | 4            | 7            |
| Recommended maximum antenna elevation for airborne survey, m                                                  | 0.3 (practical limit is 0.5m) | 0.6                                  | 0.6           | 1            | 2            | 3            |
| Minimum size of detectable objects under the surface from the recommended altitude, cm                        | 7                             | 10                                   | 10            | 20           | 35           | 50           |
| Minimum size of "deep" detectable objects from the recommended altitude, cm                                   | 11 at 0.5m                    | 26 at 2m                             | 26 at 2m      | 50 at 4m     | 100 at 8m    | 180 at 15m   |
| Minimum diameter of detectable linear non-conductive objects like an empty plastic pipe, cm                   | 5                             | 10                                   | 10            | 17           | 33           | 50           |
| Minimum diameter of detectable linear conductive objects like a metal pipe or a water-filled plastic pipe, cm | 5                             | 8                                    | 8             | 13           | 27           | 40           |



| GPR model                                                                       | Zond Aero 1000 | Zond Aero 600 & MALÅ<br>GeoDrone 600 | Zond Aero 500 | Zond Aero LF | Zond Aero LF | Zond Aero LF |
|---------------------------------------------------------------------------------|----------------|--------------------------------------|---------------|--------------|--------------|--------------|
| <b>Applications</b>                                                             |                |                                      |               |              |              |              |
| Small object search                                                             | ✓              | ✓                                    | ✓             |              |              |              |
| Glaciology, snow/ice thickness profiling                                        | ✓              | ✓                                    | ✓             | ✓            | ✓            | ✓            |
| Geological stratigraphy • subsurface stratigraphy • structure • bedrock surface |                | ✓                                    | ✓             | ✓            | ✓            | ✓            |
| Geotechnical surveys • cavity search • sinkhole search                          |                | ✓                                    | ✓             | ✓            | ✓            | ✓            |
| Utility Search • cables • water & sewage pipes • gas pipes • oil pipes          | ✓              | ✓                                    | ✓             | ✓            |              |              |
| Underground infrastructure mapping                                              | ✓              | ✓                                    | ✓             | ✓            |              |              |
| Archaeology • artifacts • hidden structures • stratigraphy • foundations        | ✓              | ✓                                    | ✓             | ✓            |              |              |
| Archaeology • caves • tombs • tunnels                                           |                |                                      |               | ✓            | ✓            | ✓            |
| Forensics archaeology                                                           | ✓              | ✓                                    | ✓             | ✓            |              |              |
| Freshwater Bathymetry                                                           |                |                                      |               | ✓            | ✓            | ✓            |
| Mining & Quarrying • rocks • fractures • faults • joints                        |                |                                      |               | ✓            | ✓            | ✓            |



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# Customized Extended Arm



Length depends upon the weight of the sensor &  
drone payload capacity

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# Full Mission Control Magnetometers



|                                                                                            | SPH Engineering<br>MagNIMBUS                                                  | SENSYS<br>MagDrone R1                      | SENSYS<br>MagDrone R3                          | SENSYS<br>MagDrone R4                          | Geometrics<br>MagArrow                                   |
|--------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|--------------------------------------------|------------------------------------------------|------------------------------------------------|----------------------------------------------------------|
| Sensor                                                                                     | 1x or 2x<br>QuSpin QTFM G2<br>laser-pumped<br>rubidium total field<br>sensors | 1x<br>FGM3D/75<br>fluxgate triaxial sensor | 2x<br>FGM3D/75<br>fluxgate triaxial<br>sensors | 5x<br>FGM3D/75<br>fluxgate triaxial<br>sensors | 2x<br>MFAM<br>laser-pumped cesium<br>total field sensors |
| Sensor sensitivity                                                                         | Very high                                                                     | High                                       | High                                           | High                                           | Very high                                                |
| <b>Recommended applications</b>                                                            |                                                                               |                                            |                                                |                                                |                                                          |
| Geology & mineral exploration                                                              | ✓                                                                             |                                            |                                                |                                                | ✓                                                        |
| Environmental (abandoned wells,<br>pipes, storage tanks, etc.)                             | ✓                                                                             | ✓                                          | ✓                                              | ✓                                              | ✓                                                        |
| Searching for UXO (unexploded<br>ordnance)                                                 | ✓                                                                             | ✓                                          | ✓                                              | ✓                                              |                                                          |
| Locating buried infrastructure (metal<br>pipes, shielded cables, and cables<br>under load) | ✓                                                                             | ✓                                          | ✓                                              | ✓                                              | ✓                                                        |
| Tramp metal and lost GET (ground<br>engagement tools), pieces of metal<br>detection        | ✓                                                                             | ✓                                          | ✓                                              | ✓                                              |                                                          |
| Archaeology                                                                                | ✓                                                                             | ✓                                          | ✓                                              | ✓                                              |                                                          |

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# Full Mission Control Magnetometers



**SENSYS MagDrone R3** magnetometer



**SENSYS MagDrone R4** magnetometer

# Integrated Bathymetry with Echo Sounder



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# Integrated Bathymetry with Echo Sounder



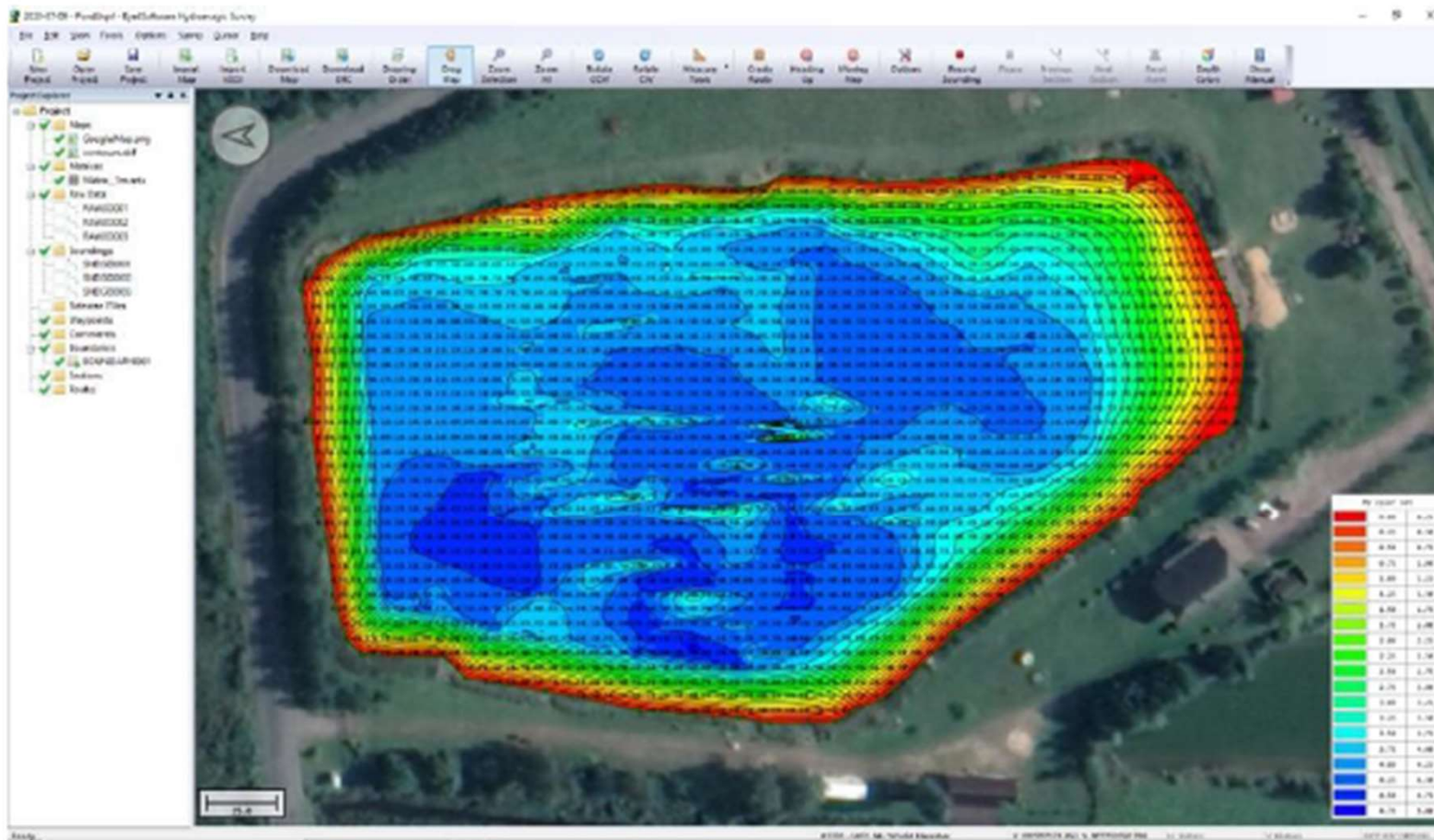
| Sensor                                                                                            | ECT 400S                                         | ECT D052S                                       | ECT D024S                                        |
|---------------------------------------------------------------------------------------------------|--------------------------------------------------|-------------------------------------------------|--------------------------------------------------|
| Type                                                                                              | Single frequency                                 | Dual frequency                                  | Dual frequency                                   |
| Acoustic frequency                                                                                | 450 kHz                                          | 50/200 kHz                                      | 200 kHz / 450 KHz                                |
| Measurement range*                                                                                | 0.15 ... 100m                                    | 1.0 ... 200m (50 kHz)<br>0.5 ... 200m (200 kHz) | 0.5 ~ 200m (200 kHz)<br>0.15 ~100m (450 kHz))    |
| Beam width<br>Conical (-3dB)                                                                      | 5°                                               | 27° / 7° (50 kHz/200 kHz)                       | 10° / 5°                                         |
| Weight of the echo sounder (in the air)                                                           | 275g                                             | 460g                                            | 280g                                             |
| Weight of all components in the air (echo sounder, SkyHub, altimeter, housing, cables, mountings) | 1.6 kg (light housing)<br>2.6 kg (heavy housing) | 2.7 kg                                          | 1.6 kg (light housing)<br>2.6 kg (heavy housing) |

# Integrated Bathymetry with Echo Sounder



In addition to depth measurements, water temperature and tilt angles of the sensor are logged.

Example of a colored depth map, data processed in the HydroMagic software:



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# CBRN Drones: Full Mission Control



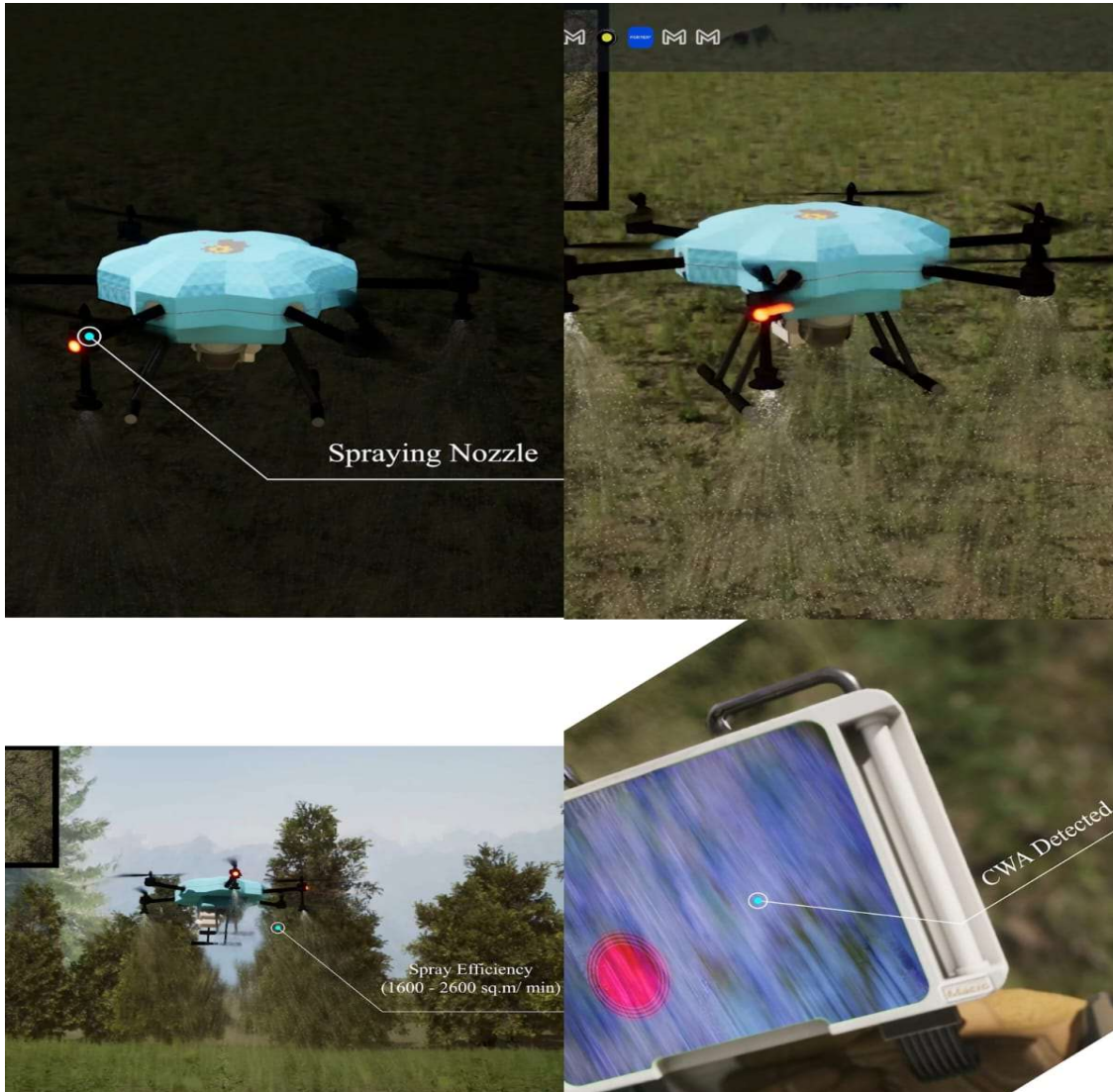
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# CBRN Drones: Full Mission Control



Follow the link to see video

<https://youtu.be/dyGy4pnRcql>



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# The Offboard Compute Advantage

Redefining mission flexibility and onboard  
efficiency.

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# The GCS: Your Control Center



## Offboard Computational Power

All heavy computation (e.g., target tracking, dynamic replanning, motion detection) is performed at the Ground Control Station (GCS), not on the drone.

## Onboard Simplicity

The drone requires no onboard computer. This radically reduces airborne weight, lowers power draw, and dramatically increases reliability by minimizing points of failure.



## GCS: Full Mission Control



**Live Video:** Real-time monitoring and control feed.



**Mission Planning:** Autonomous waypoint upload and management.



**Manual Override:** Full joystick control for tactical maneuvers.



**Target Tracking:** Real-time automated tracking of selected targets.



**Dynamic Replan:** Update mission parameters in real-time from the GCS.

# Autonomous & Advanced Ops

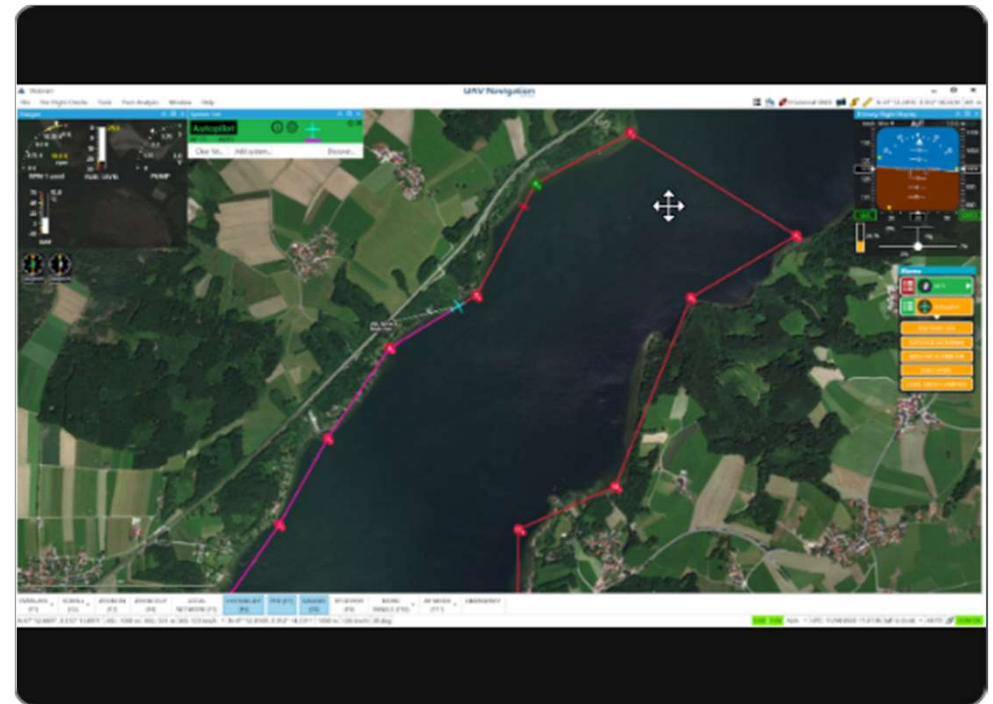


## Flight Modes

- Loiter, Stabilize, Guided, plus fully autonomous VTOL takeoff, landing, and waypoint navigation.

## Advanced Capabilities

- Automated motion detection.
- Persistent target tracking via GCS.
- Dynamic flight plan adjustment (retasking).



# Reliability: Failsafe Systems



## Comms Failure

Automatically executes a Return-to-Home (RTH) protocol upon loss of communication.



## Low Battery

Triggers a safe and immediate auto-land protocol at critical battery levels.



## Safe Altitudes

All failsafe behaviors are executed at user-configurable safe altitudes to avoid obstacles.

# The Multi-Mission Advantage



## Payload & Portability

High payload capacity (8kg) on a compact, transportable wheelbase.



## Persistent ISR

Long endurance (45 mins) combined with a high-quality, long-range video feed.



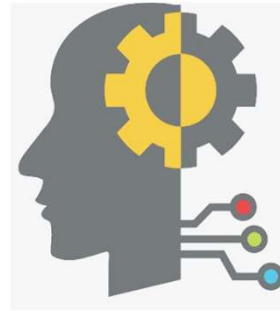
## Task Agnostic

Offboard compute makes the drone a flexible tool, adaptable for any mission.

# Drone with Payload : Sensors & Instruments



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**Pune, Maharashtra**  
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